Energy accounting for a SEFI-inspired vibration harvester

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Purpose and finding

Test whether an identity-organized controller can improve net energy capture from changing vibrations after tuning and controller costs are counted. This is an engineering interpretation inspired by SEFI resonance and stability ideas. It is not derived from the proposed physical field equations and introduces no new energy source.

Finding: adaptive tuning can improve the modeled result on some changing inputs, but loses energy on the weak input under the assumed power budget. The SEFI-labeled event controller is exactly equivalent to the matched conventional event controller. No SEFI-specific gain or publishable novelty has been demonstrated.

A concrete engineering model

The source is an externally prescribed sinusoidal force on a linear mass-spring-damper oscillator: m x_ddot + (c_m + c_e) x_dot + k(t) x = F(t). Electrical extraction is represented by ideal damping, with P_e = c_e x_dot squared. This is extracted mechanical power at an idealized transducer, not measured DC power delivered by hardware.

Stored mechanical energy is E = (m x_dot squared + k x squared)/2. Between stiffness updates, dE/dt = F x_dot - (c_m+c_e) x_dot squared. At an instantaneous stiffness update, delta E = x squared delta k / 2. That energy change must be accounted for; tuning is not free.

Positive stiffness-change work is supplied by a modeled actuator at 80% efficiency. Released spring energy is discarded without electrical recovery. Each stiffness change also costs 0.1 microjoule. Adaptive controllers consume an assumed 10 microwatts throughout the 30-second run. These values are illustrative assumptions, not measured specifications.

Net metric: extracted electrical energy minus actuator supply energy, switching costs, and controller energy. Source work is reported separately in the machine-readable ledger. Negative net energy means the modeled controller/actuator budget exceeds extraction; it does not imply negative harvested power.

Scale and intended use

This is a low-power sensor-scale study: millimeter motion and microjoule-scale net results. It supports neither utility-scale generation nor claims of free energy. A physical generator, rectifier, storage interface, sensing system, and tuner remain unspecified.

Methods and fair comparison

Quantity	Assumption
Mass / mechanical damping	0.02 kg / 0.01 N s/m
Electrical damping	0.02 N s/m normally; 0.08 above a 3 mm amplitude proxy
Tuning range	5 to 15 Hz, through k = m (2 pi f) squared
Actuator update limit	At most 5 N/m change per 0.05-second control interval
Initial state / adaptive tuning	Zero displacement and velocity; initial tuning 10 Hz
Excitation	0.002 N peak normally; 0.0004 N for weak chirp
Numerics	Classical RK4, 1 ms; repeated at 0.5 ms for refinement

The displacement rule is a reactive damping heuristic, not a proof of constraint satisfaction. Every run reports peak travel and limit violations. Fixed runs have constant normal electrical damping; no run reached the 3 mm travel threshold.

Frequency is estimated from positive zero crossings of the observed source force, using interpolated crossing times. No controller is given the future source frequency. A practical instrument must obtain an equivalent signal and pay its sensing cost. Noise, sensor bias, coupling back to the source, and electrical conversion losses are not modeled.

Fixed: select the highest-net-energy setting from 5, 7.5, 10, 12.5, and 15 Hz after each scenario is simulated. This is an offline, scenario-informed comparator; it favors the fixed baseline and is not a deployable causal selection rule.

Adaptive: track estimated resonance when stiffness error exceeds 1%, with the shared slew limit and damping heuristic. **Conventional event:** use a 5% deadband. **SEFI candidate:** apply that same event rule with Origin as device constants, Authorship as observations, Sovereignty as operating limits, and Warp as accepted stiffness updates. These are engineering label assignments, not the four-coordinate physical dynamics.

The event and SEFI candidate implementations share the same update mathematics. Their exact equality is a deliberate ablation. It prevents attributing ordinary tuning behavior to the SEFI names. The deadbands and power assumptions were selected illustratively, not fitted to demonstrate a benefit.

Four deterministic inputs

Steady: 10 Hz for 30 seconds. Steps: 5 Hz for 10 seconds, 15 Hz for 10 seconds, then 8 Hz. Chirp: a continuous frequency ramp from 5 to 15 Hz. Weak chirp: the same ramp with one-fifth force amplitude. Phase is continuous. There is no random ensemble; these runs do not establish statistical robustness.

Simulation results

30-second input	Best fixed	Adaptive	Event / SEFI
steady	1244.4	944.4	944.4
steps	386.5	629.3	369.5
chirp	49.2	673.4	136.9
weak chirp	2.0	-276.9	-287.8

All table values are net microjoules. Event and SEFI results agree exactly. Machine-readable files include gross extraction, source work, actuator work, released spring energy, controller consumption, mechanical loss, terminal storage, and energy residual.

At 1 ms, the maximum absolute mechanical energy-ledger residual is 2.25e-09 J. Halving the timestep changes net energy by at most 1.56 microjoules across the listed cases and controllers. Detailed refinement results are supplied rather than treating ledger closure as proof of controller convergence.

The identity-labeled candidate is not superior to its matched conventional baseline. Improvements over fixed tuning, where present, are attributable to ordinary resonance tracking in this model. On weak excitation, the assumed controller consumes more energy than the system extracts. Hardware overhead, power gating, passive adaptation, and uncertainty could change the comparison.

A responsible next experiment

Ask an experienced harvesting laboratory to select an existing low-voltage, small-displacement testbed, calibrate its electrical and mechanical transfer model, and measure tuner and sensing consumption. Start with known vibration inputs and fixed, adaptive, and matched event controllers. Include source work and stored-energy changes, not only peak voltage.

The primary outcome should be net delivered electrical energy at the same source conditions and travel limits. Record current and voltage simultaneously, mechanical input force and velocity (or a calibrated equivalent), control supply, tuner supply, and final storage. Repeat trials with varied phase, noise, source spectra, and component tolerances. Instrumentation costs and externally powered test equipment must be clearly separated from the device budget.

A SEFI-specific hypothesis must change a testable update rule or prediction beyond generic constraints, deadband control, or regularization. If a matched conventional controller reproduces the result, report that equivalence. Do not claim a new physical mechanism from a renamed controller.

Readiness

Appropriate now: a concise request for technical criticism or referral, with code available. Not established: a novel controller, hardware feasibility, improved net delivered power, a new physical energy interaction, or readiness for journal submission. The wider SEFI manuscripts are provenance, not evidence of device performance.

Recommended expert route

First technical-fit contact: Professor Alper Erturk, Smart Structures and Dynamical Systems Laboratory, Georgia Tech. Official profiles describe theoretical and experimental work in vibration, nonlinear energy harvesting, smart structures, and programmable systems. Public email: alper.erturk@me.gatech.edu. This is a topic-fit recommendation, not an assurance of availability, review, or collaboration.

Alternative: Professor Lei Zuo, University of Michigan. Official faculty and laboratory pages list energy harvesting, vibration/dynamics and control, and marine renewable energy. Public lab contact: leizuo@umich.edu. Approach one group first with a focused request; this package is not a journal submission.

Possible later publication: Journal of Intelligent Material Systems and Structures covers theoretical and experimental smart/adaptive structures research. Its current instructions require novel work for research articles and experimental proof for applied papers. This exploratory benchmark does not meet those claims of novelty or validation yet. An expert should first assess whether a distinct controller or device contribution is possible.

Sources checked October 18, 2026

Georgia Tech research profile: https://me.gatech.edu/faculty/erturk

Georgia Tech public contact: https://energy.gatech.edu/people/alper-erturk

University of Michigan faculty profile: https://me.engin.umich.edu/people/faculty/lei-zuo/

Michigan laboratory team and public contact: https://marine.engin.umich.edu/our-team

Journal submission requirements: https://journals.sagepub.com/author-instructions/jim

Zhu, Tudor and Beeby, 2010, frequency-range review, university repository: https://eprints.soton.ac.uk/268354/

Energy-optimized self-adaptive harvesting, 2019, related prior research: https://doi.org/10.3390/mi10010044

The literature check establishes that resonance tuning and accounting for tuning energy already exist. It is not an exhaustive priority search. The 2019 paper was located through its bibliographic record; its full text was not successfully retrieved.

Provenance and reproducibility

SEFI inspiration belongs to Jason D. Dutton, whose earliest theory source is the illustrated SEFI book. This oscillator and control study is an additional engineering construction prepared with AI assistance on October 18, 2026. It does not validate the book's particle, emergent-spacetime, or energy-source claims. No experiment was performed. No institutional affiliation, funding, or endorsement is claimed. Author review and independent technical validation remain necessary.

Run study.py with Python 3; it uses only the standard library. Parameters, source signals, update policies, and ledger are explicit in code. results.json, results.csv, and convergence.json reproduce the tabulated outputs. package.py builds this brief using ReportLab. Existing manuscripts remain separate; disclose any substantive overlap before future publication.